Critical Thinking Module 4: Hospital Medication-Delivery Robot

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For this assignment, I developed an interactive Hospital Medication-Delivery Robot Route Planner that models a hospital floor as a finite, weighted search space. An intelligent-search problem must identify where the agent begins, where the goal is, which actions are legal, which states are reachable, and how the agent can estimate its distance from the goal (Colorado State University Global, n.d.). In the program, each state is represented by an immutable (row, column) coordinate; the user selects one hospital department as the initial state and another as the goal state; the robot can move up, right, down, or left; walls are impassable; and normal corridors, busy corridors, and sanitation work zones have traversal costs of 1, 3, and 6, respectively. The program uses a custom implementation that mirrors the SimpleAI problem structure through the actions, result, `is_goal`, `cost`, and `heuristic` operations (SimpleAI Team, n.d.). I selected A* graph search because it compares candidate routes with the evaluation function $f(n) = g(n) + h(n)$, where $g(n)$ is the exact path cost already accumulated and $h(n)$ estimates the cost remaining to reach the goal. In this way, A* combines the cost awareness of Uniform-Cost Search with the goal direction provided by Greedy Best-First Search (CS188 Spring 2014, 2014). The heuristic is the Manhattan distance to the destination multiplied by the minimum legal movement cost of one. It is admissible because it ignores walls and higher-cost terrain, so it cannot overestimate the least possible remaining route cost, and it is consistent because one legal movement changes the Manhattan distance by no more than one while costing at least one unit. For this finite map, A* is complete because all movement costs are positive and repeated-state control prevents the search from cycling indefinitely; with the admissible and consistent heuristic, it also returns a minimum-cost route (CS188 Spring 2014, 2014; Hart et al., 1968). A major advantage of A* is that it considers both distance and terrain cost, allowing the robot to avoid a geometrically short but expensive route through a sanitation work zone. Its main disadvantage is that A* is not especially space-efficient because it retains frontier nodes and the best-known cost for each discovered state, which can require substantial memory in larger search spaces. Manhattan distance can also provide weak guidance when walls force long detours. In the default Pharmacy-to-Emergency demonstration, the program reaches the goal in 38 movements with a total route cost of 44 units and displays the route map, action sequence, and the $g(n)$, $h(n)$, $f(n)$ and values for each step.

References

Colorado State University Global (n.d.). [Interactive Lecture]. *Lecture 4: Intelligent Search Methods.* CSC510: Foundations of Artificial Intelligence. Canvas

CS188 Spring 2014. (2014). *CS188 SP14 lecture 3 informed search* [Video]. YouTube. https://www.youtube.com/watch?v=8pTjoFiICg8

Hart, P. E., Nilsson, N. J., & Raphael, B. (1968). A formal basis for the heuristic determination of minimum cost paths. *IEEE Transactions on Systems Science and Cybernetics, 4*(2), 100–107. https://doi.org/10.1109/TSSC.1968.300136

SimpleAI Team. (n.d.). *Search algorithms*. SimpleAI documentation. https://simpleai.readthedocs.io/en/latest/search_problems.html